

Identifiability of Deuteron Tensor Polarization with One and Two L/R/U/D Polarimeters

DPOLAR analysis note

Generated from repository conventions; numerical tables are reproducible analysis products

Abstract

This note separates three questions that are often conflated in discussions of deuteron polarimetry: whether a tensor component appears in the cross section, whether the detector geometry provides an independent response to it, and whether that response survives profiling of luminosity and detector nuisance parameters. The analysis uses the Cartesian spin-1 convention already used by the repository. An ideal single L/R/U/D ring has one exact tensor-geometry null direction, p_{xy} , when absolute normalization is known. With unknown global normalization its common-mode p_{zz} response is also lost at one polar angle. Multiple polar angles can recover p_{zz} only when their acceptance-averaged A_{zz} responses differ and share a constrained normalization. A second station increases structural rank only when known spin transport rotates one station's null space away from the other's. Around an axial vertical p_{yy} state, one infinitesimal tilt is visible at first order and the orthogonal tilt is invisible in a perfectly cardinal four-arm geometry. These results support an after-SRC polarimeter as a direct monitor of the tensor polarization delivered by the accelerator chain, while making the scientific case for a second station conditional on a specific spin-transport or contamination measurement.

1 Scope, authority, and limitations

The source of truth for polarization normalization is the repository's Cartesian density-matrix convention. The source of truth for the measured reaction response is the repository's polarized d - p elastic-scattering formula. The current tabulation contains iT_{11} , T_{20} , T_{21} , and T_{22} as functions of deuteron center-of-mass scattering angle. The legacy C++ scalar count path presently consumes only T_{20} and T_{22} ; the full response requires all four observables.

All rank claims below are analytic unless explicitly labeled as generated results. Scenario-specific singular values, Fisher eigenvalues, uncertainties, and figures must be inserted from the machine-readable output of the analysis executable. Placeholders in this note are deliberate and must not be read as physics forecasts.

The current repository does not contain a complete, sourceable, element-by-element spin-transfer map from the proposed after-SRC station to the proposed pre-SAMURAI station. Rotation scans are therefore generic sensitivity studies, not predictions of the RIBF beam line.

2 Spin-1 tensor degrees of freedom

In the ordered S_z basis, define the Hermitian Cartesian spin-1 tensor operators

$$Q_{ij} = \frac{3}{2}(S_i S_j + S_j S_i) - 2\delta_{ij}I, \quad p_i = \text{Tr}(\rho S_i), \quad p_{ij} = \text{Tr}(\rho Q_{ij}). \quad (1)$$

The density matrix is expanded as

$$\rho = \frac{I}{3} + \frac{1}{2} \sum_i p_i S_i + \frac{1}{9} \sum_i p_{ii} Q_{ii} + \frac{2}{9} \sum_{i < j} p_{ij} Q_{ij}. \quad (2)$$

The tensor polarization is the real symmetric traceless matrix

$$\mathbf{P} = \begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{xy} & p_{yy} & p_{yz} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix}, \quad p_{xx} + p_{yy} + p_{zz} = 0. \quad (3)$$

It therefore has five independent degrees of freedom. For a state diagonal in magnetic substates about an axis a ,

$$p_a = N_{+1} - N_{-1}, \quad p_{aa} = N_{+1} + N_{-1} - 2N_0 = 1 - 3N_0. \quad (4)$$

For zero vector polarization the axial tensor range is $-2 \leq p_{aa} \leq 1$. The legacy scalar scan range $0 \leq p \leq 1$ is a scenario choice and must not be mistaken for the physical tensor range.

2.1 Orthonormal numerical basis and physical reporting basis

A well-conditioned internal basis is orthonormal under the Frobenius inner product, $\langle A, B \rangle = \text{Tr}(A^\top B)$:

$$E_1 = \frac{1}{\sqrt{2}} \text{diag}(1, -1, 0), \quad E_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad (5)$$

$$E_4 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad E_5 = \frac{1}{\sqrt{6}} \text{diag}(-1, -1, 2). \quad (6)$$

Writing $q_a = \text{Tr}(\mathbf{P} E_a)$ gives

$$\mathbf{q} = \begin{pmatrix} (p_{xx} - p_{yy})/\sqrt{2} \\ \sqrt{2} p_{xy} \\ \sqrt{2} p_{xz} \\ \sqrt{2} p_{yz} \\ \sqrt{3/2} p_{zz} \end{pmatrix}. \quad (7)$$

The numerical basis must not replace the experiment-facing reporting basis

$$\mathbf{r} = (p_{xx} - p_{yy}, p_{xy}, p_{xz}, p_{yz}, p_{zz})^\top. \quad (8)$$

The inverse relations are

$$p_{xx} - p_{yy} = \sqrt{2} q_1, \quad (p_{xy}, p_{xz}, p_{yz}) = \frac{1}{\sqrt{2}} (q_2, q_3, q_4), \quad p_{zz} = \sqrt{\frac{2}{3}} q_5, \quad (9)$$

with $p_{xx} = [(p_{xx} - p_{yy}) - p_{zz}]/2$ and $p_{yy} = [-(p_{xx} - p_{yy}) - p_{zz}]/2$.

3 Polarized elastic-scattering response

In the local incident-beam frame used by the repository,

$$\begin{aligned} \frac{\sigma(\theta, \phi)}{\sigma_0(\theta)} = & 1 + \frac{3}{2}(p_x \sin \phi + p_y \cos \phi)A_y(\theta) \\ & + \frac{2}{3}(p_{xz} \cos \phi - p_{yz} \sin \phi)A_{xz}(\theta) \\ & + \frac{1}{6}[(p_{xx} - p_{yy}) \cos 2\phi - 2p_{xy} \sin 2\phi][A_{xx}(\theta) - A_{yy}(\theta)] \\ & + \frac{1}{2}p_{zz}A_{zz}(\theta). \end{aligned} \quad (10)$$

The repository's spherical-to-Cartesian conversion is

$$A_y = \frac{2}{\sqrt{3}}iT_{11}, \quad A_{xz} = -\sqrt{3}T_{21}, \quad A_{xx} - A_{yy} = 2\sqrt{3}T_{22}, \quad A_{zz} = \sqrt{2}T_{20}. \quad (11)$$

Equations (10) and (11) reproduce the maintained scalar weights

$$f_{p_{zz}} = 1 + \frac{\sqrt{2}}{2}T_{20}p_{zz}, \quad (12)$$

$$f_{LR/UD} = 1 \mp \frac{\sqrt{3}}{2}T_{22}p_{yy} - \frac{\sqrt{2}}{4}T_{20}p_{yy} \quad (13)$$

for a longitudinal axial tensor and a vertical axial tensor, respectively.

An older displayed closed-form LR/UD inversion contains a factor-of-two error in its A_{zz} denominator. From Eq. (10) and the maintained sector weights, the cardinal-ring result is

$$R_{LRUD} = \frac{N_L + N_R - N_U - N_D}{N_L + N_R + N_U + N_D} = -\frac{p_{yy}(A_{xx} - A_{yy})}{4 - p_{yy}A_{zz}}. \quad (14)$$

The forward-model likelihood follows the maintained sector weights and is not replaced by the obsolete analytic denominator.

The signs of the U/D first harmonics in an older expanded set of paper equations disagree with Eq. (10). This analysis adopts the general equation and the maintained response implementation: L, U, R, D are at $\phi = (0, \pi/2, \pi, 3\pi/2)$, so $U - D$ has a negative $p_{yz}A_{xz}$ sign. Changing the experimental azimuth convention requires changing both channel metadata and interpretation, not only plot labels.

4 Acceptance-integrated channel means

For detector channel k , the expected Poisson mean is

$$\mu_k(\mathbf{q}, \boldsymbol{\eta}) = \mathcal{L} \epsilon_k \int_{\Omega_k} d\Omega \sigma_0(\theta) \left[1 + \sum_{a=1}^5 c_a(\theta, \phi) q_a + \sum_{v=1}^3 d_v(\theta, \phi) p_v \right], \quad (15)$$

where $d\Omega = \sin \theta d\theta d\phi$, $\mathcal{L}$ contains beam dose and target areal density, ϵ_k is a channel efficiency, and the coefficient functions are obtained from Eq. (10) and the basis conversion. Define the unpolarized integral and tensor response coefficients

$$b_k = \epsilon_k \int_{\Omega_k} d\Omega \sigma_0(\theta), \quad (16)$$

$$H_{ka} = \epsilon_k \int_{\Omega_k} d\Omega \sigma_0(\theta) c_a(\theta, \phi). \quad (17)$$

Then tensor-only means are linear,

$$\boldsymbol{\mu} = \mathcal{L}(\mathbf{b} + H\mathbf{q}). \quad (18)$$

The physical target is an integral over the actual detector face. Center-angle evaluation is used only for the analytic ring check. The numerical study in this note instead follows the maintained finite rectangular top-hat acceptance, including analytic integration of the azimuthal harmonics across each opening; the approximation is stated in every output.

4.1 Legacy acceptance caution

The maintained scalar C++ count integrator presently multiplies a midpoint sum in θ_{cm} by a constant $\Delta\phi$ but omits the $\sin\theta_{\text{cm}}$ solid-angle factor. A separate coverage-map path does include $\sin\theta d\theta d\phi$, but that map is not used by scalar count inference. Consequently:

1. existing scalar numerical outputs remain regression anchors and must not be changed silently;
2. the present general response deliberately preserves the legacy $d\theta d\phi$ measure while adding finite-azimuth harmonics, so its absolute rate forecasts are provisional;
3. absolute rates from the legacy and corrected measures must not be mixed;
4. a future exact circular-face integration with $\sin\theta$ must be a separately named acceptance mode and reported with before-and-after regression tests.

5 Response matrix and local identifiability

For parameters θ_a , define

$$J_{ka} = \frac{\partial\mu_k}{\partial\theta_a}. \quad (19)$$

For tensor coordinates, $J_{ka} = \mathcal{L}H_{ka}$. If the luminosity nuisance is parameterized as $\lambda = \log \mathcal{L}$, then $\partial\mu_k/\partial\lambda = \mu_k$. A log parameter avoids scale-dependent finite differences and enforces positive luminosity.

For the response in the orthonormal basis of Eq. (7), with no additional column scaling, compute

$$J = U\Sigma V^\top. \quad (20)$$

The reported output must contain all singular values and right-singular vectors, not only an integer rank. A singular value is counted when

$$\sigma_i > \tau, \quad \tau = \max(\tau_{\text{abs}}, \tau_{\text{rel}}\sigma_1). \quad (21)$$

A suitable default after documented column scaling is $\tau_{\text{rel}} = 10^{-10}$ and $\tau_{\text{abs}} = 10^{-12}$, accompanied by a threshold scan. Rank conclusions must be stable over at least two orders of magnitude around the chosen relative threshold. Every null or near-null right singular vector must be converted from $\mathbf{q}$ to the physical basis (8) and normalized with a stated sign convention.

6 Analytic sanity check: one ideal L/R/U/D ring

Let $B = A_{xx} - A_{yy}$ and use the physical tensor coordinates (8). At a fixed polar angle, the polarization part of the response row at azimuth ϕ is

$$h(\phi) = \left(\frac{B}{6} \cos 2\phi, -\frac{B}{3} \sin 2\phi, \frac{2A_{xz}}{3} \cos \phi, -\frac{2A_{xz}}{3} \sin \phi, \frac{A_{zz}}{2} \right). \quad (22)$$

For L, U, R, D this becomes

$$H_{\text{ring}} = C \begin{pmatrix} B/6 & 0 & 2A_{xz}/3 & 0 & A_{zz}/2 \\ -B/6 & 0 & 0 & -2A_{xz}/3 & A_{zz}/2 \\ B/6 & 0 & -2A_{xz}/3 & 0 & A_{zz}/2 \\ -B/6 & 0 & 0 & 2A_{xz}/3 & A_{zz}/2 \end{pmatrix}, \quad (23)$$

where C is the common unpolarized acceptance integral.

6.1 Geometry null and measured harmonics

Equation (23) gives, without hard-coded null vectors,

$$L - R \propto \frac{4}{3} p_{xz} A_{xz}, \quad (24)$$

$$U - D \propto -\frac{4}{3} p_{yz} A_{xz}, \quad (25)$$

$$(L + R) - (U + D) \propto \frac{2}{3} (p_{xx} - p_{yy}) B. \quad (26)$$

Because $\sin 2\phi = 0$ at all four cardinal azimuths, p_{xy} is an exact geometry null. If B , A_{xz} , and A_{zz} are nonzero and absolute normalization is known, the tensor response rank is four.

6.2 Common mode and normalization

The p_{zz} column is proportional to $(1, 1, 1, 1)^T$. It is an independent observable if absolute normalization is externally known. If a global normalization is free, the normalization derivative is the same common mode at the unpolarized point. After profiling it, the effective tensor rank of a single ideal ring is three: $(p_{xx} - p_{yy})$, p_{xz} , and p_{yz} remain, while p_{xy} is geometric-null and p_{zz} is normalization-null. Away from the unpolarized point the local null is a state-dependent combination of scale and tensor common mode, but the physical conclusion is unchanged.

This distinction is the simplest example of why raw response rank and profiled polarization rank must both be reported.

7 Multiple polar angles within one station

For ring g at polar angle θ_g , the common-mode response is $A_{zz}(\theta_g)/2$. With a single shared luminosity, two angular settings separate p_{zz} from luminosity if their normalized response vectors are not collinear. In the point-ring limit this requires

$$A_{zz}(\theta_1) \neq A_{zz}(\theta_2), \quad (27)$$

and in the real detector it requires unequal acceptance-weighted averages,

$$\frac{H_{p_{zz}}^{(1)}}{b_1} \neq \frac{H_{p_{zz}}^{(2)}}{b_2}. \quad (28)$$

If each angular group has its own unconstrained normalization, this recovery is lost. Multiple θ values can also separate vector A_y from tensor A_{xz} first harmonics when their angular dependences are not proportional. They cannot recover p_{xy} from perfectly cardinal, azimuth-symmetric arms, because the $\sin 2\phi$ integral remains zero at every polar angle.

The production study evaluates proton and deuteron singles separately and also evaluates proton–deuteron coincidences. A union of inclusive singles and coincidences is deliberately not assigned an independent-Poisson rank because shared events and shared efficiencies require a consistent exclusive likelihood or covariance model.

Table 1: Three-hour, unpolarized-point response results for the planned compact geometry approximation, with one shared profiled global normalization. Singular values are derivatives in counts per unit orthonormal tensor coordinate; the relative/absolute raw-rank thresholds are $10^{-10}/10^{-12}$.

Configuration	Raw/profiled rank	Spectrum and dominant nulls
One θ , four azimuths	4/3 analytic	p_{xy} geometry null; p_{zz} /scale profiled null
Two- θ proton singles	4/4	$(1.8603, 1.8603, 1.4074, 1.0658, 0) \times 10^5$; p_{xy} null
Deuteron singles, branches summed	4/3	$(5.7183, 2.9425, 2.9425, 0.3166, 0) \times 10^5$; common mode profiled
Deuteron branches, energy resolved	4/4	$(5.0582, 2.0818, 2.0818, 1.7427, 0) \times 10^5$; p_{xy} null
Forward/backward coincidences	4/4	$(1.3879, 1.3879, 1.0460, 0.8852, 0) \times 10^5$; p_{xy} null
Per-angle normalized proton singles	4/3	p_{zz} recovery is lost when each angle has a free scale

8 Poisson Fisher information and nuisance profiling

For independent Poisson channels and full parameter vector $\boldsymbol{\theta} = (\mathbf{q}, \boldsymbol{\eta})$,

$$F_{ab} = \sum_k \frac{1}{\mu_k} \frac{\partial \mu_k}{\partial \theta_a} \frac{\partial \mu_k}{\partial \theta_b}. \quad (29)$$

Partitioning polarization and nuisance parameters gives

$$F = \begin{pmatrix} F_{qq} & F_{q\eta} \\ F_{\eta q} & F_{\eta\eta} \end{pmatrix}. \quad (30)$$

The effective polarization information after nuisance profiling is

$$F_{\text{eff}} = F_{qq} - F_{q\eta} F_{\eta\eta}^+ F_{\eta q}, \quad (31)$$

where $+$ denotes a tolerance-controlled pseudoinverse. Gaussian calibration priors add their precision matrix before the Schur complement. The output must include the full Fisher matrix, F_{eff} , its eigensystem, covariance F_{eff}^+ , and identifiable-subspace uncertainties.

At minimum the nuisance model must support a global luminosity scale. Useful stress tests add per-angle scales, four relative sector efficiencies, and analyzing-power modes. Completely free per-channel efficiencies absorb every polarization response and eliminate identifiability; calibration priors or symmetry constraints are therefore part of the measurement definition.

The present scalar code propagates pointwise T_{20}/T_{22} uncertainties in quadrature rather than fitting analyzing-power nuisance parameters or a table covariance. That legacy uncertainty band is not equivalent to Eq. (31) and must be labeled separately.

9 Spin transport in the five-dimensional tensor space

For $R \in SO(3)$,

$$\mathbf{p}_2 = R\mathbf{p}_1, \quad \mathbf{P}_2 = R\mathbf{P}_1 R^\top. \quad (32)$$

The induced five-dimensional representation in the orthonormal basis is

$$T_{ab}^{(2)}(R) = \text{Tr} \left[E_a^\top R E_b R^\top \right], \quad \mathbf{q}_2 = T^{(2)}(R) \mathbf{q}_1. \quad (33)$$

Because the basis is orthonormal and rotations preserve the Frobenius inner product, $T^{(2)}(R)$ is orthogonal. Therefore

$$\text{Tr}(\mathbf{P}_2^2) = \text{Tr}(\mathbf{P}_1^2), \quad \|\mathbf{q}_2\|_2 = \|\mathbf{q}_1\|_2. \quad (34)$$

Tests must cover identity, rotations about each Cartesian axis, generic Euler rotations, composition, and norm preservation. This layer depends only on $SO(3)$ and accepts future Thomas–BMT or transfer-map rotations without embedding a particular magnet lattice.

10 One station versus two stations

Let $\mathbf{q}_0$ be the tensor at a common reference point and T_s the known rank–2 transport to station s . The station response to the reference tensor is $J_s T_s$, so

$$J_{\text{combined}} = \begin{pmatrix} J_1 T_1 \\ J_2 T_2 \end{pmatrix}. \quad (35)$$

The central geometric statement is

$$\boxed{\ker J_{\text{combined}} = \ker(J_1 T_1) \cap \ker(J_2 T_2)}. \quad (36)$$

A second station increases structural rank if and only if this intersection has smaller dimension than the first station’s null space. Equivalently, at least one direction invisible to station 1 must have a nonzero projection into the row space of $J_2 T_2$.

Two identical stations with $T_2 = T_1$ have aligned null spaces. Doubling their exposure improves nonzero Fisher eigenvalues and statistical precision but does not change rank. A rotation may still fail to help if it leaves the null subspace invariant, if the relevant analyzing power vanishes, or if newly generated response is absorbed by independent station normalizations or efficiencies. Thus “more counts” and “more identifiable dimensions” are different statements.

With station-specific nuisances, the correct comparison is between profiled Fisher matrices, not only stacked polarization Jacobians. Shared luminosity, common analyzing-power calibration, and independent station efficiencies must be encoded according to the actual experiment.

11 Generic rotation sweep and null-space complementarity

For a relative rotation $R_a(\alpha)$ about $a = x, y, z$, scan $0 \leq \alpha \leq \pi$. At each point report

1. raw and profiled effective rank;
2. all singular values and the smallest nonzero singular value;
3. condition number restricted to the identifiable subspace;
4. Fisher determinant when full rank, otherwise pseudo-determinant;
5. covariance eigenvalues in the identifiable subspace;
6. principal angles between transported station null spaces.

If N_1 and N_2 are orthonormal bases for the two null spaces, the cosines of their principal angles are the singular values of $N_1^T N_2$. A principal angle of zero identifies a shared null direction. The dimension of the exact intersection must agree with Eq. (36).

Extrema in a rotation scan describe mathematical complementarity. They are not preferred accelerator settings until a real transport map, acceptance model, beam phase space, and nuisance constraints are supplied.

Table 2: Generic two-station rotation scan for identical planned proton-channel geometry at the two stations, a shared global normalization, and a 2° scan grid. These are mathematical sensitivity rotations, not an RIBF transfer-map prediction.

Rotation family	Unchanged-degeneracy angles	Maximum complementarity	Profiled rank
$R_x(\alpha)$	$0^\circ, 180^\circ$	90° : $\sigma_{\min} = 1.6414 \times 10^5$	five for sampled 2° – 178°
$R_y(\alpha)$	$0^\circ, 180^\circ$	90° : $\sigma_{\min} = 1.6414 \times 10^5$	five for sampled 2° – 178°
$R_z(\alpha)$	$0^\circ, 90^\circ, 180^\circ$	$44^\circ/46^\circ$: $\sigma_{\min} = 1.0470 \times 10^5$	five except listed angles

12 Local analysis around nominal pure p_{yy}

An axial tensor with magnitude P_T and unit symmetry axis $\hat{n}$ is

$$\mathbf{P}(P_T, \hat{n}) = \frac{3}{2}P_T\hat{n}\hat{n}^\top - \frac{1}{2}P_TI. \quad (37)$$

This normalization is fixed by the repository convention: if $\hat{n} = \hat{y}$, then $p_{yy} = P_T$ and $p_{xx} = p_{zz} = -P_T/2$.

Parameterize small tilts about the vertical direction as

$$\hat{n}(\alpha, \beta) = \left(\alpha, \sqrt{1 - \alpha^2 - \beta^2}, \beta \right), \quad (38)$$

where α tilts the axis toward $+x$ and β tilts it toward $+z$. At $\alpha = \beta = 0$,

$$\frac{\partial \mathbf{P}}{\partial P_T} = \text{diag}(-1/2, 1, -1/2), \quad (39)$$

$$\frac{\partial \mathbf{P}}{\partial \alpha} = \frac{3}{2}P_T(\hat{x}\hat{y}^\top + \hat{y}\hat{x}^\top), \quad (40)$$

$$\frac{\partial \mathbf{P}}{\partial \beta} = \frac{3}{2}P_T(\hat{y}\hat{z}^\top + \hat{z}\hat{y}^\top). \quad (41)$$

Thus the x -tilt generates p_{xy} at first order and is exactly invisible to an ideal cardinal L/R/U/D ring. The z -tilt generates p_{yz} and appears in the $U - D$ first harmonic when $A_{xz} \neq 0$. Multiple polar angles do not lift the exact p_{xy} null if every arm remains centered and azimuthally symmetric. Non-cardinal azimuths, known acceptance asymmetry, or a transported second station can make that tilt visible; uncontrolled asymmetry can instead mimic it.

The local Fisher analysis in (P_T, α, β) must include normalization and efficiency nuisances. At exactly $P_T = 0$, orientation is undefined and both tilt derivatives vanish, so an axis ellipse is meaningful only around a nonzero nominal tensor state. A second station helps when its transport maps the upstream p_{xy} perturbation into a tensor harmonic observed by its local geometry. Identity transport between identical stations cannot recover the missing tilt.

Unwanted tensor contamination should be reported by projecting the five-dimensional covariance onto the three-dimensional tangent space of the axial model and its two-dimensional orthogonal complement. This distinguishes uncertainty in magnitude/axis from non-axial contamination rather than forcing all deviations into a tilt.

13 Detector scenarios and unresolved geometry values

Several repository artifacts preserve different historical detector values. The maintained default scalar configuration uses proton laboratory angles 53.4° and 11.2° , whereas an older paper table

Table 3: Nominal vertical-state local statistical sensitivity at $P_T = 0.8$ and three hours for the planned two- θ proton channels, profiling one shared global normalization. Analyzing-power, background, alignment, and efficiency systematics are not included.

Configuration	$\sigma(P_T)$	Tilt covariance/ellipse	Contamination eigenmodes
One station	0.00378	α unidentifiable; $\sigma(\beta) = 0.00180$ rad (0.103°)	full-tensor p_{xy} null
Two stations, $R_y(20^\circ)$	0.00267	$\sigma(\alpha) = 0.00721$ rad, $\sigma(\beta) = 0.00180$ rad	no structural tensor null

uses 55.9° and 11.3°. Distances and active sizes also differ. The compact common detector configuration currently uses 53.4° at 205 mm, 11.2° at 190 mm, and a deuteron channel at 20.9° and 140 mm; its detector response and several mechanical inputs remain provisional.

These values must not be averaged or silently “corrected.” Every generated result must record the exact source configuration, resolved channel manifest, angle convention, distance reference, active-face dimensions, beam energy, table checksums, and acceptance algorithm. Results for the historical paper geometry and the current compact geometry are different named scenarios. The after-SRC and pre-SAMURAI compact configurations inherit the same common physics channel layout at present; their different chamber interfaces alone do not create a different polarization response matrix. The CLI selection named **current** denotes the planned two-angle proton singles response only. Deuteron singles and coincidence samples are separate named studies; inclusive samples are not combined as independent Poisson bins.

14 Implication for an after-SRC polarimeter

The relevant conceptual chain is

$$\begin{aligned} & \text{PIS} \rightarrow \text{Wien filter} \rightarrow \text{AVF} \rightarrow \text{RRC} \rightarrow \text{SRC} \\ & \rightarrow \boxed{\text{after-SRC polarimeter}} \rightarrow \text{downstream beam line} \rightarrow \text{SAMURAI}. \end{aligned} \quad (42)$$

The source and Wien filter prepare the intended spin mode. The cyclotrons and extraction system transport that state. A post-SRC polarimeter independently anchors the tensor polarization actually delivered by the complete acceleration chain at final energy.

This argument does not require, and must not be phrased as, an expectation of large SRC depolarization. The defensible statement is: good polarization preservation is expected under proper accelerator operation, while a post-SRC measurement independently verifies the delivered tensor polarization and constrains polarization-related systematics.

For the immediate nominal pure- p_{yy} program, one after-SRC station can be sufficient for the primary task of magnitude and stability monitoring if the axial-state assumption, angular response, relative efficiencies, and relevant normalization are controlled. That statement is narrower than arbitrary five-component tensor reconstruction. A second station is scientifically justified when it addresses an additional quantified question, for example:

1. measuring a transport-induced tensor-axis rotation predicted by a known beam-line map;
2. lifting the first-order p_{xy} tilt/contamination null through a demonstrably complementary rotation;
3. separating coherent spin transport from non-axial contamination or ensemble depolarization;
4. providing a downstream polarization anchor when intervening beam-line elements introduce a systematic uncertainty relevant at SAMURAI.

The justification must be expressed through the reduction of $\ker(J_1 T_1) \cap \ker(J_2 T_2)$ or through a specified improvement in profiled uncertainty, not merely through doubled event statistics.

15 Required machine-readable outputs and figures

For every scenario, the analysis product must include the parameter basis and conversion matrix, channel metadata, unpolarized vector $\mathbf{b}$, response matrix H , full Jacobian, nuisance definitions and priors, column scales, singular values and vectors, rank thresholds, null vectors in both bases, Fisher matrices, profiled Fisher matrix, covariance/pseudoinverse, uncertainty eigenvectors, spin rotation and $T^{(2)}(R)$, and software/configuration provenance. JSON is the canonical structured summary and CSV stores matrices and scans without lossy formatting.

The following figures are referenced only after generation from those products.

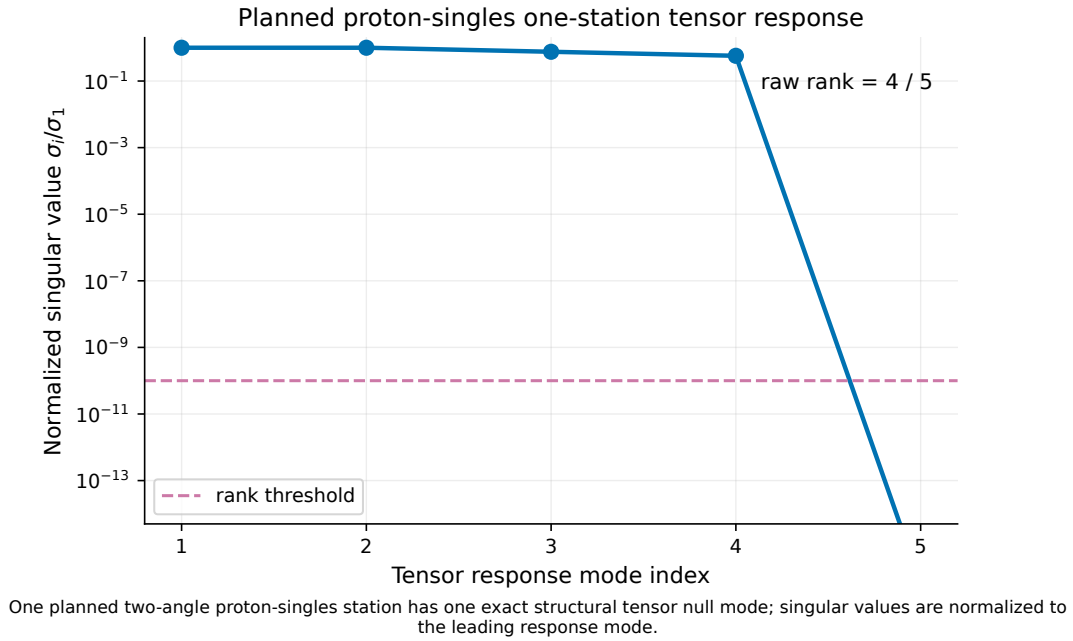


Figure 1: Singular-value spectrum for one station. The caption in the generated figure states the orthonormal basis convention and rank threshold.

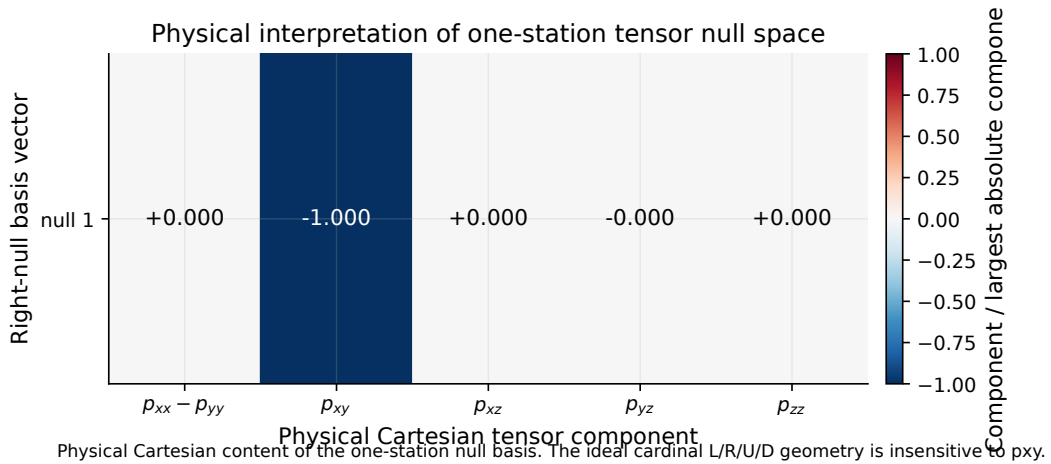
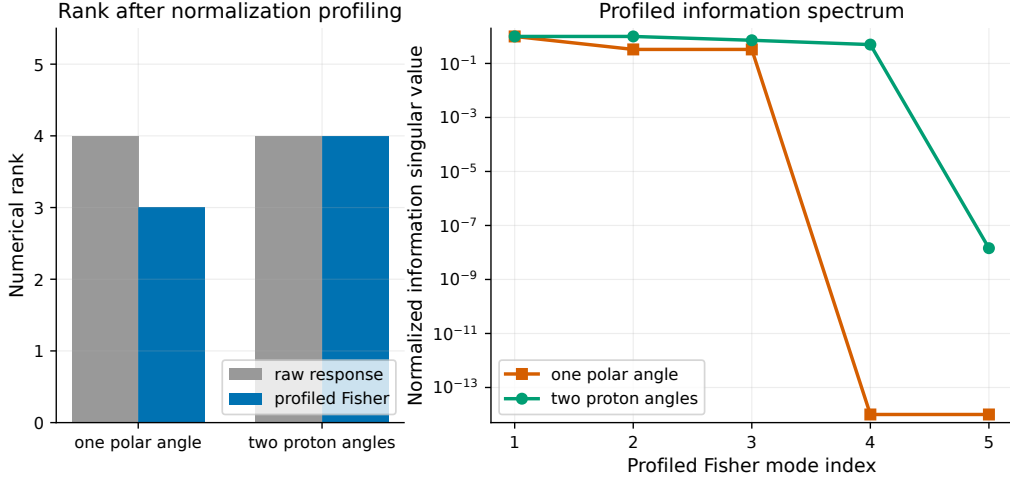


Figure 2: Tensor null and near-null directions translated into the physical Cartesian reporting basis.



A second polar-angle group changes the analyzing-power dependence and restores one polarization mode after global-normalization profiling.

Figure 3: One-polar-angle versus multiple-polar-angle response, separating structural rank from conditioning.

16 Validation requirements

The tensor layer must test symmetric/traceless round trips, Cartesian/internal basis round trips, $\mathbf{P}' = \mathbf{R}\mathbf{P}\mathbf{R}^T$, identity rotation, generic rotation composition, and norm preservation. The ideal ring test must obtain the p_{xy} null and the three non-common harmonics from the constructed matrix, not from hard-coded conclusions. A normalization test must show the single-angle profiled-rank loss and conditional recovery with two genuinely different angular responses. Two-station tests must show that identical transport preserves rank, that exposure improves precision without changing rank, that a selected rotation reduces the null-space intersection, and that a direct numerical intersection agrees with Eq. (36).

Existing scalar p_{zz} and p_{yy} inference values are regression anchors. Any change caused by the solid-angle correction, interpolation method, or geometry resolution requires a focused test and an explicit before/after record; it must not enter as an incidental side effect of the rank analysis.

17 Conclusions

One ideal cardinal L/R/U/D ring supplies three normalization-independent tensor harmonics: diagonal transverse anisotropy, p_{xz} , and p_{yz} . It has an exact p_{xy} geometry null. Its p_{zz} common mode is measurable with known absolute normalization and degenerate with an unknown single-ring scale. Multiple polar angles can recover p_{zz} when their acceptance-averaged A_{zz} responses differ under a shared constrained normalization, but cannot recover p_{xy} while exact cardinal azimuth symmetry remains.

A second station is not automatically a new tensor measurement. Its value is determined by the transported null-space intersection in Eq. (36) and by nuisance-aware Fisher information. For the nominal vertical axial state, magnitude and the tilt generating p_{yz} are locally accessible to one ideal station, while the orthogonal tilt generating p_{xy} is first-order invisible. A known complementary spin rotation can make that missing direction observable at a second station.

The after-SRC instrument is therefore well motivated as the direct final-energy anchor for the intended pure- p_{yy} beam. A second station should be pursued when a real transport map

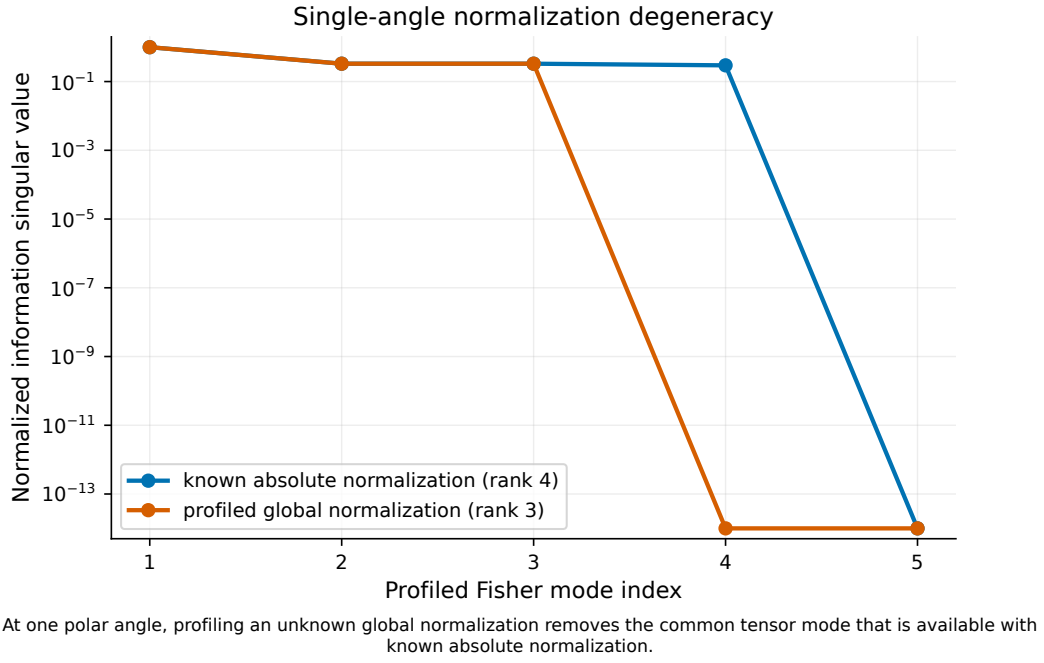


Figure 4: Known-normalization and profiled-normalization comparison. Loss of a common mode must not be labeled a detector-geometry null.

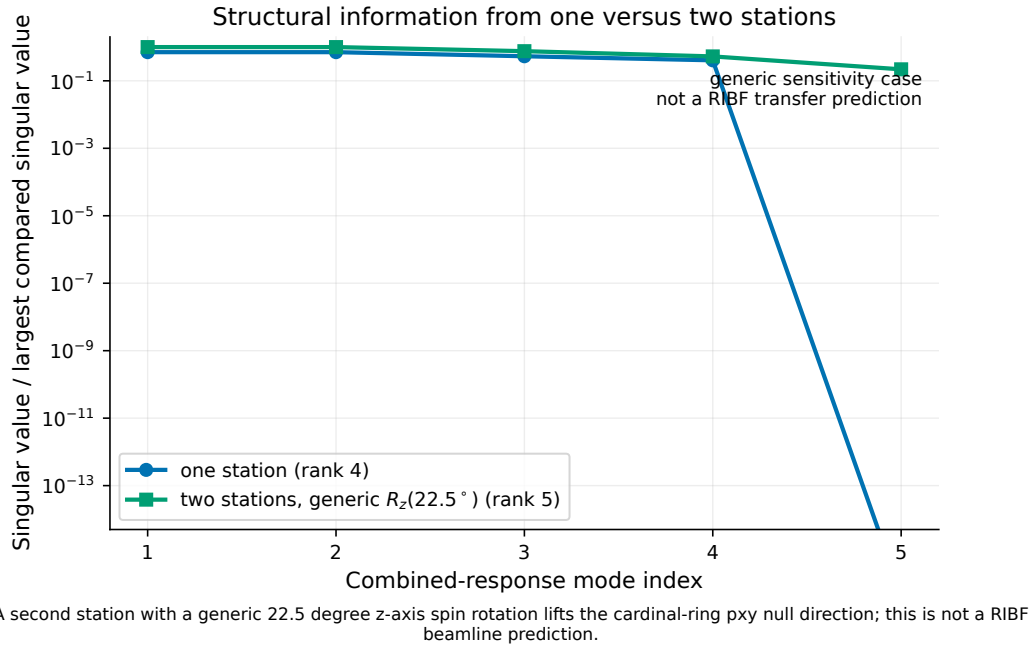


Figure 5: One-station and two-station singular spectra for a stated relative spin rotation.

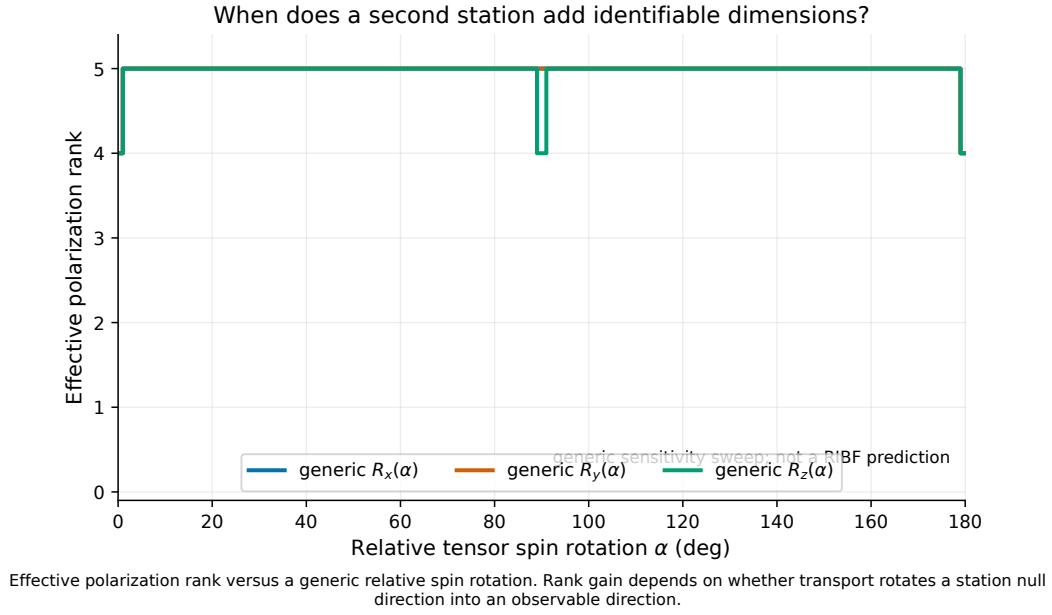


Figure 6: Effective rank versus generic relative rotation. This is a sensitivity study, not a beam-line prediction.

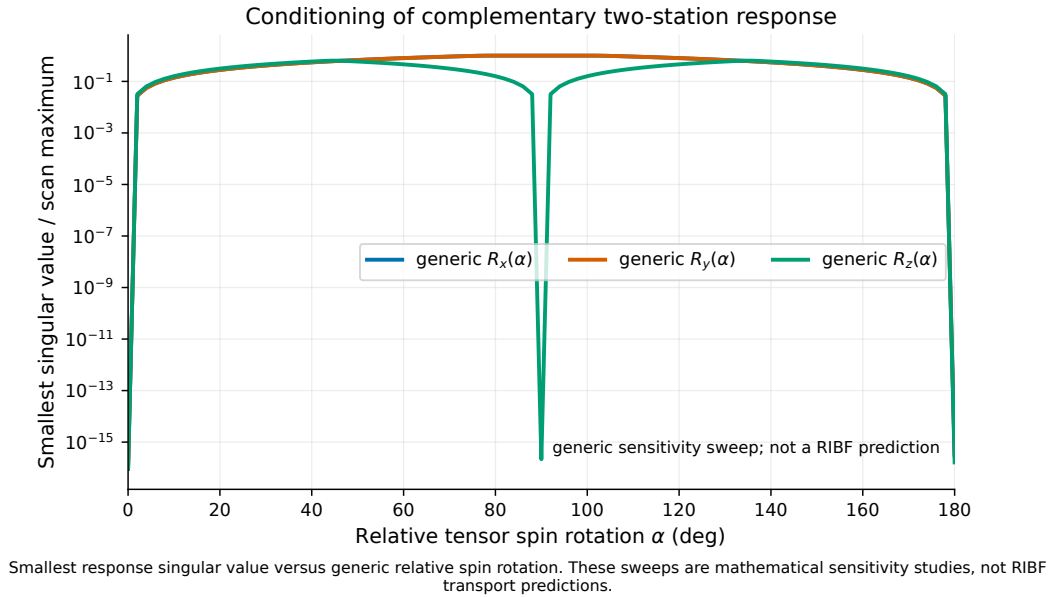
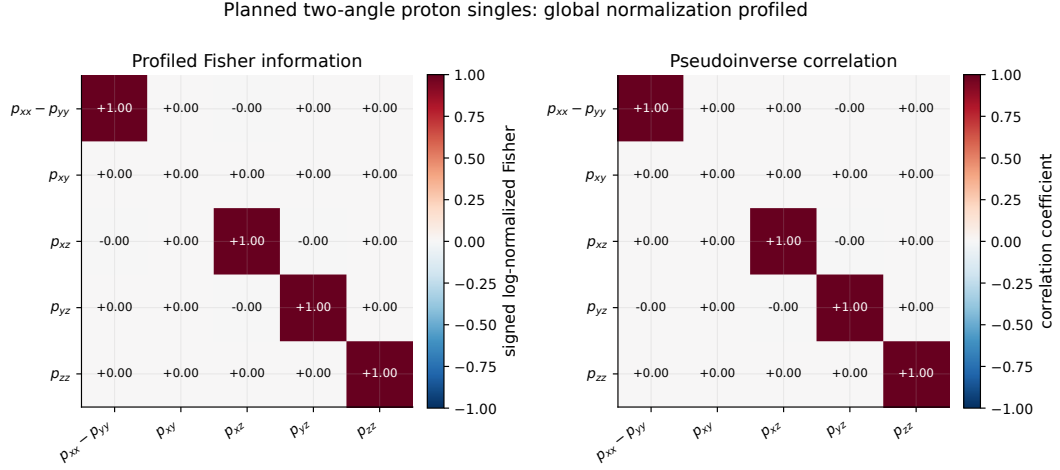


Figure 7: Smallest singular value and threshold versus relative rotation angle.



Profiled Fisher information and pseudoinverse correlation for the planned two-angle proton-singles response with global normalization treated as a nuisance.

Figure 8: Profiled Fisher/correlation matrix for the planned two-angle proton-singles scenario and nuisance model.

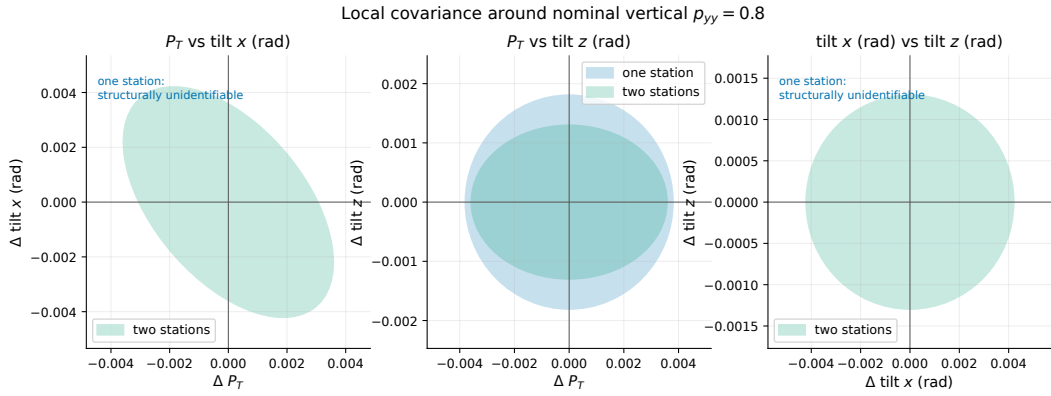


Figure 9: Profiled local covariance for (P_T, α, β) around the stated nonzero nominal vertical tensor state.

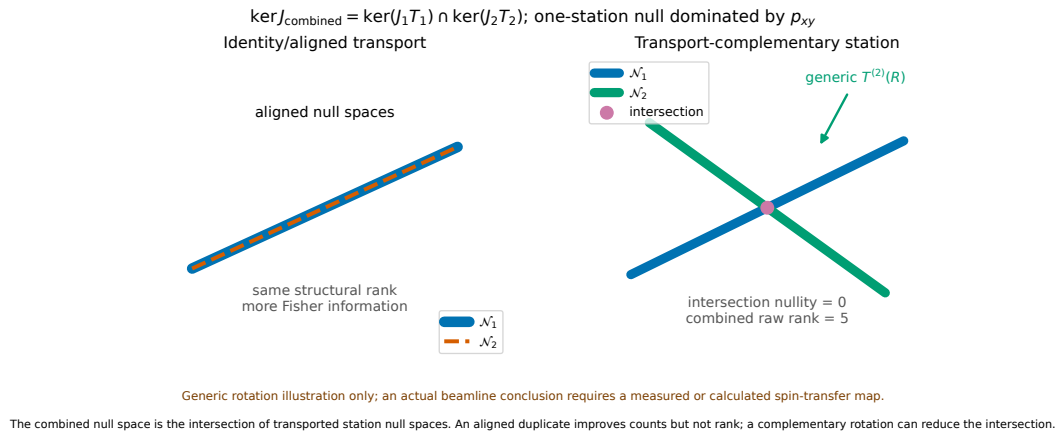


Figure 10: Null-space interpretation of Eq. (36); overlapping null directions remain unobservable after combining stations.

and a quantified null-space or uncertainty reduction show that it answers a scientifically distinct downstream question.